

Topic 10: Chemistry of the environment - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Air, water, climate and industrial chemistry

Questions in this answer PDF: 20

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2020 May/June Paper 41 Question 3

Main topic: Topic 10: Chemistry of the environment | Marks: 17 | Answer source: 0620_s20_ms_41.pdf

Extra topic(s): Topic 7: Acids, bases and salts, Topic 6: Chemical reactions, Topic 9: Metals

Answer points

3(a)

- zinc blende

3(b)(i)

- reaction is reversible
- rate of forward reaction = rate of reverse reaction

3(b)(ii)

- vanadium(V) oxide

3(b)(iii)

- increases the rate of reaction

3(b)(iv)

- particles have more energy (E)
- rate of collisions increase
- a higher proportion of particles have energy greater than activation energy ($E > E_A$)

3(b)(v)

- exothermic

3(b)(vi)

- Mr of $\text{SO}_3 = 80$
- 40%
- x
- =

3(c)

- anhydrous copper(II) sulfate
- carbon

3(d)(i)

- not all (C-C) bonds are single

3(d)(ii)

- $\text{C}_3\text{H}_7\text{OH} \rightarrow \text{C}_3\text{H}_6 + \text{H}_2\text{O}$
- C_3H_6
- rest of the equation

3(d)(iii)

- propene

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2020 May/June Paper 42 Question 3

Main topic: Topic 10: Chemistry of the environment | Marks: 16 | Answer source: 0620_s20_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

3(a)(i)

- heat in air

3(a)(ii)

- $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$ (1)

- 450°C (1)

- vanadium(V) oxide (1)

3(a)(iii)

- rate of forward reaction and rate of backward reaction are equal (1)

- concentrations of reactants and products are constant (1)

3(b)

- concentrated sulfuric acid

3(c)

- $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4$

3(d)

- $\text{Cu} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$

3(e)

- purple to colourless

3(f)

- $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$

3(g)(i)

- barium nitrate / barium chloride

3(g)(ii)

- $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$

- formulae (1) state symbols (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2020 May/June Paper 43 Question 2

Main topic: Topic 10: Chemistry of the environment | Marks: 18 | Answer source: 0620_s20_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

2(a)(i)

- reversible reaction

2(a)(ii)

- hydrocarbons (reacting with steam)

2(b)

- [increasing pressure] increases yield

- [increasing temperature] decreases yield

2(c)

- (particles) have more energy OR (particles) move faster

- more collisions per second OR greater collision rate

- more (of the) particles OR collisions have sufficient energy / activation energy to react OR a greater percentage /

- proportion / fraction of collisions are successful

2(d)(i)

- N / NH_3

- change in oxidation state of N from -3 to +2 / increase in oxidation number / gain in oxygen / loss of electrons

2(d)(ii)

2(d)(iii)

- (it could react with rain water to) form nitric acid / acid rain

2(d)(iv)

- (Mr of HNO_3 =) 63 (1)

- 20 (1)

- 5 (1)

- 120 (dm^3) (1)

2(e)

- -

- horizontal product energy line at lower energy level than reactant
- -
- label of product
- -
- correct direction of vertical arrow - arrow must start level with reactant energy and finish level with product level
- and one arrow head ONLY

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2020 May/June Paper 43 Question 4

Main topic: Topic 10: Chemistry of the environment | Marks: 8 | Answer source: 0620_s20_ms_43.pdf

Extra topic(s): Topic 12: Experimental techniques and chemical analysis

Answer points

4(a)

- (filtration:) remove solids from water / remove insoluble substances
- (chlorination:) sterilises / kill microbes / prevent illness

4(b)(i)

- white to blue

4(b)(ii)

- higher boiling point / greater than 100°C

4(c)(i)

- more than one spot

4(c)(ii)

- 0.8 (circled)

4(c)(iii)

- use a locating agent

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2020 Oct/Nov Paper 41 Question 4

Main topic: Topic 10: Chemistry of the environment | Marks: 10 | Answer source: 0620_w20_ms_41.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds, Topic 7: Acids, bases and salts

Answer points

4(a)

4(b)(i)

- fractional (1)
- distillation (1)

4(b)(ii)

- (different) boiling point

4(c)

- 2 double bonds (1)
- whole molecule correct (2 pairs of lone pairs on each O) (1)

4(d)

- increase in (concentrations of) carbon dioxide
- (carbon dioxide is) greenhouse gas/greenhouse effect
- contributes to climate change/global warming

4(e)

- photosynthesis

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2020 Oct/Nov Paper 43 Question 3

Main topic: Topic 10: Chemistry of the environment | Marks: 20 | Answer source: 0620_w20_ms_43.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds

Answer points

3(a)

- triple bond (1)
- diagram completely correct (1)

3(b)(i)

- METHOD 1
- liquid air (1)
- fractional distillation (1)
- METHOD 2
- hydrogen
- burns in air (to remove the oxygen and then scrub out the carbon dioxide)

3(b)(ii)

- (pressure) 200 atmospheres (1)
- (temperature) 450 °C (1)
- iron catalyst (1)
- $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (1)
- equilibrium / reversible (1)

3(c)(i)

- substance that speeds up a reaction / increases rate (1)
- unchanged (chemically) at the end
- OR not used up
- OR lowers activation energy (1)

3(c)(ii)

- gain of oxygen / loss of hydrogen / electron loss / increase in oxidation state (oxidation number)

3(c)(iii)

- effect on the rate of the
- forward reaction
- effect on the equilibrium
- yield of $\text{NO}(\text{g})$
- increase (1)
- decrease (1)
- increase (1)
- decrease (1)

3(d)

- $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$
- all formulae (1)
- equation fully correct(1)

3(e)

- (Mr of NH_4NO_3 =) 80 (1)
- 35% (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2021 Oct/Nov Paper 41 Question 4

Main topic: Topic 10: Chemistry of the environment | Marks: 12 | Answer source: 0620_w21_ms_41.pdf

Extra topic(s): Topic 6: Chemical reactions

Answer points

4(a)

- $4\text{FeCuS} + 13\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 4\text{CuO} + 8\text{SO}_2$
- Fe_2O_3 and CuO as a product (1)
- Equation fully correct (1)

4(b)(i)

- rate of forward reaction = rate of reverse reaction (1)

- concentration of reactants and products are constant (1)

4(b)(ii)

- 450 °C (1)
- 1-2 atm (1)

4(b)(iii)

- vanadium(V) oxide

4(b)(iv)

- increase rate of reaction

4(b)(v)

- equilibrium shifts to left hand side (1)
- forward reaction is exothermic (1)

4(c)

- $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 6\text{C} + 6\text{H}_2\text{O}$
- H_2O (1)
- balance (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2021 Oct/Nov Paper 43 Question 1

Main topic: Topic 10: Chemistry of the environment | Marks: 9 | Answer source: 0620_w21_ms_43.pdf

Extra topic(s): Topic 11: Organic chemistry, Topic 9: Metals

Answer points

1(a)

- oxygen

1(b)

- bauxite

1(c)

- sulfur dioxide

1(d)

- ethanol

1(e)

- sodium chloride

1(f)

- ammonia

1(g)

- carbon dioxide

1(h)

- carbon monoxide

1(i)

- ammonia

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2022 May/June Paper 41 Question 4

Main topic: Topic 10: Chemistry of the environment | Marks: 10 | Answer source: 0620_s22_ms_41.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

4(a)(i)

- $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$

4(a)(ii)

- (temperature) 450 °C (1)
- (pressure) 1-2 atmosphere(s) (1)
- vanadium(V) oxide catalyst (1)

- $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ (1)

4(a)(iii)

- $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$

4(a)(iv)

- water

4(b)(i)

- 2 bonding pairs as one dot and cross each (1)

- 2 lone pairs on S (and no additional electrons on Hs) to complete the outer shell on S and both Hs (1)

4(b)(ii)

- $2\text{H}_2\text{S} + \text{SO}_2 \rightarrow 3\text{S} + 2\text{H}_2\text{O}$

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2022 Oct/Nov Paper 42 Question 3

Main topic: Topic 10: Chemistry of the environment | Marks: 16 | Answer source: 0620_w22_ms_42.pdf

Extra topic(s): Topic 9: Metals, Topic 7: Acids, bases and salts, Topic 6: Chemical reactions

Answer points

3(a)

- zinc blende

3(b)

- strong heating in air / roasting in air

3(c)

- contact

3(d)(i)

- 450 (°C) (1)

- 1-2 (atm) (1)

3(d)(ii)

- vanadium(V) oxide

3(d)(iii)

- the rate of forward reaction equals (the rate of the) reverse reaction (1)

- concentrations of reactants and products are constant (1)

3(d)(iv)

- increased temperature:

- (position of) equilibrium moves to left-hand side (1)

- reaction is exothermic (1)

- increased pressure:

- (position of) equilibrium moves to right-hand side (1)

- more (gaseous) moles / molecules on left-hand side (1)

3(d)(v)

- rate increases and particles have more energy (1)

- more collisions (between particles) occur per second / per unit time (1)

- more of the particles / collisions have energy equal to or above the activation energy (1)

- or

- more of the particles / collisions have sufficient energy to react

- or

- a higher percentage / proportion / fraction of collisions (of particles) are successful / have energy equal to or above

- activation energy

3(e)

- ammonium sulfate

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2022 Oct/Nov Paper 43 Question 5

Answer points

5(a)(i)

- 450 (1)
- 1-2 (1)
- vanadium (V) oxide (1)

5(a)(ii)

- $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$

5(a)(iii)

- water

5(b)

- carbon

5(c)(i)

- $2.5 \rightarrow 10^{-3} / 0.0025$ (1)
- $1.25 \rightarrow 10^{-3} / 0.00125$ (1)
- 20 (1)

5(c)(ii)

5(d)(i)

- lilac flame

5(d)(ii)

- Any Two from
- -
- solid disappears
- -
- blue solution
- -
- bubbles / effervescence / fizzing

5(d)(iii)

- white precipitate

5(e)

- $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
- BaSO_4 on the right (1)
- ONLY Ba^{2+} and SO_4^{2-} on the left (1)
- state symbols (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2023 Feb/March Paper 42 Question 1

Answer points

1(a)

- carbon dioxide

1(b)

- 78 (%)

1(c)

- calcium oxide

1(d)

- catalytic converter

1(e)(i)

- CH_4

1(e)(ii)

- lowest relative molecular mass

1(f)

- toxic

1(g)

- glucose
- oxygen

1(h)

- two dot-and-cross double bonds
- two pairs of non-bonding electrons on O and zero non-bonding electrons on C

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2023 Feb/March Paper 42 Question 3

Main topic: Topic 10: Chemistry of the environment | Marks: 18 | Answer source: 0620_m23_ms_42.pdf

Extra topic(s): Topic 6: Chemical reactions, Topic 5: Chemical energetics

Answer points**3(a)**

- nitrogen: air (1)
- hydrogen: methane (1)

3(b)(i)

- enthalpy change

3(b)(ii)

- (the value of) ΔH is negative

3(b)(iii)

- 450 (1)
- 20 000 (1)
- iron (1)

3(b)(iv)

- one mark for each of
- decreases
- decreases
- decreases
- no change

3(b)(v)

- kinetic energy of particles increases (1)
- frequency of collisions between particles increases (1)
- higher percentage / proportion / fraction of collisions / particles have energy greater than / equal to activation energy (1)
- or
- more of the collisions / particles have energy greater than / equal to activation energy

3(c)(i)

- H_2SO_4

3(c)(ii)

- fertiliser
- February/March 2023

3(c)(iii)

- Mr of $(\text{NH}_4)_2\text{SO}_4 = 132$ (1)
- $2 \rightarrow 14 = 28$ and $\%N = 100 \rightarrow 28 / 132 = 21.2\%$ (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2023 May/June Paper 43 Question 6

Main topic: Topic 10: Chemistry of the environment | Marks: 15 | Answer source: 0620_s23_ms_43.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points**6(a)(i)**

- air

6(a)(ii)

- methane

6(a)(iii)

- 450 (°C)
- 200 (atm)

6(a)(iv)

- iron

6(a)(v)

- (a substance which) increases the rate of a reaction
- remains unchanged at the end of the reaction

6(b)(i)

- temperature change:
- low(er) rate (of reaction)
- pressure change:
- (position of) equilibrium shifts to the left hand side/ towards reactants

6(b)(ii)

- $4\text{NO} + 3\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$

6(c)(i)

- $\text{CuCO}_3 + 2\text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{CO}_2 + \text{H}_2\text{O}$
- $\text{Cu}(\text{NO}_3)_2$
- correct equation

6(c)(ii)

- undissolved solid
- effervescence stops on addition of more copper(II) carbonate

6(d)(iii)

- copper(II) oxide or
- copper(II) hydroxide

Easy explanation / reminder

- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2023 Oct/Nov Paper 41 Question 1

Main topic: Topic 10: Chemistry of the environment | Marks: 6 | Answer source: 0620_w23_ms_41.pdf

Extra topic(s): Topic 11: Organic chemistry, Topic 7: Acids, bases and salts

Answer points**1(a)**

- sulfur dioxide

1(b)

- ammonia

1(c)

- xenon

1(d)

- oxygen

1(e)

- ethene

1(f)

- ammonia

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2023 Oct/Nov Paper 43 Question 3

Main topic: Topic 10: Chemistry of the environment | Marks: 21 | Answer source: 0620_w23_ms_43.pdf

Extra topic(s): Topic 6: Chemical reactions, Topic 2: Atoms, elements and compounds

Answer points**3(a)(i)****3(a)(ii)**

- natural gas

3(a)(iii)

- three single bonding pairs containing one dot and one cross (1)
- two dots or two crosses on N (and no additional electrons on Hs) to complete the outer shell on N and all 3 Hs (1)

3(a)(iv)

- $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (1)
- ■ (1)
- 450 °C (1)
- 200 atmospheres / 20 000 kPa (1)
- iron (1)

3(b)(i)

- platinum

3(b)(ii)

- -3 (1)
- +2 (1)

3(b)(iii)

- oxidation AND oxidation number increases

3(b)(iv)

- increase (1)
- decrease (1)
- increase (1)
- no change (1)

3(b)(v)

- kinetic energy of particles is lower (1)
- frequency of collisions between particles is lower (1)
- fewer / lower percentage / lower proportion / lower fraction of particles have energy greater than / equal to activation energy (1)
- OR
- fewer / lower percentage / lower fraction of collisions have energy greater than / equal to activation energy (1)

3(c)

- $4\text{NO} + 3\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$

Easy explanation / reminder

- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2024 May/June Paper 42 Question 1

Main topic: Topic 10: Chemistry of the environment | Marks: 7 | Answer source: 0620_s24_ms_42.pdf

Extra topic(s): Topic 9: Metals

Answer points**1(a)(i)**

- methane

1(a)(ii)

- nitrogen dioxide

1(a)(iii)

- propene

1(a)(iv)

- helium

1(a)(v)

- carbon monoxide

1(b)

- $2\text{NO}_2 + 4\text{CO} \rightarrow \text{N}_2 + 4\text{CO}_2$
- $\text{N}_2 + \text{CO}_2$ as only products
- correct equation

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2025 Feb/March Paper 42 Question 1

Main topic: Topic 10: Chemistry of the environment | Marks: 6 | Answer source: 0620_m25_ms_42.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds

Answer points

1(a)

1(b)

1(c)

1(d)

1(e)

1(f)

- 6.02 → 1023

Easy explanation / reminder

- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

2025 May/June Paper 42 Question 1

Main topic: Topic 10: Chemistry of the environment | Marks: 8 | Answer source: 0620_s25_ms_42.pdf

Extra topic(s): Topic 8: The Periodic Table

Answer points

1(a)

- 78(%)

1(b)

1(c)

1(d)

1(e)

- any value between 25 to 35 inclusive

1(f)

1(g)

1(h)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration × volume. Then use the mole ratio from the balanced equation before converting to the final answer.

2025 May/June Paper 43 Question 4

Main topic: Topic 10: Chemistry of the environment | Marks: 13 | Answer source: 0620_s25_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 7: Acids, bases and salts

Answer points

4(a)(i)

- $4\text{FeS}_2 + 11\text{O}_2 \rightarrow 8\text{SO}_2 + 2\text{Fe}_2\text{O}_3$

4(a)(ii)

- double bond of 2 dots and 2 crosses

- 2 non-bonding pairs of dots on one oxygen atom and 2 non-bonding pairs of crosses on the other oxygen atom

4(b)(i)

- (temperature)

- 450 and °C

- (pressure)

- 2 and atm

- OR

- 200 and kPa

- vanadium(V) oxide catalyst

4(b)(ii)

- $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$

4(b)(iii)

- $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$

4(c)

- zero

- + 4

4(d)

- mol $\text{NaHCO}_3 = 4.20 / 84 = 0.05(00)$
- mol $\text{CO}_2 = 0.05(00)$
- vol $\text{CO}_2 = 0.05 \times 24\,000 = 1200$

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*